

That depends on each investor's risk aversion. We saw in the previous chapter that we can summarize mean-variance preferences by *indifference curves*. Maximizing mean-variance utility is equivalent to choosing the highest possible indifference curve. Indifference curves correspond to the individual maximizing mean-variance utility (see equation (2.4) in chapter 2 restated here):

$$\begin{aligned} \max_{\{w_i\}} E(r_p) - \frac{\gamma}{2} \text{var}(r_p), \\ \text{subject to any constraints.} \end{aligned} \quad (3.3)$$

The coefficient of risk aversion, γ , is specific to each individual. The weights, $\{w_i\}$, correspond to the risky assets in the investor's universe. Investment in the risk-free asset, if it is available, constitutes the remaining investment (all the asset weights, including the risk-free asset position, sum to one).

3.1. WITHOUT A RISK-FREE ASSET

Figure 3.8 shows the solution method for the case without a risk-free asset.²⁰ The left graph in the top row shows the indifference curves. As covered in chapter 2, one particular indifference curve represents one level of utility. The investor has

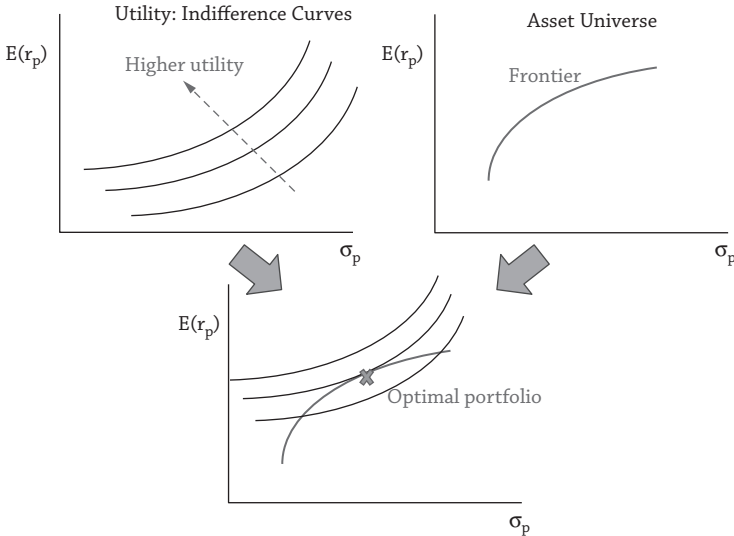


Figure 3.8

²⁰The mathematical formulation of this problem corresponds to equation (3.3) with the constraint that the weights in risky assets sum to one or that the weight in the risk-free asset is equal to zero.